CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

Slide Set 12: SVM Dual Formulation

Slides from CS 171 / EE 142

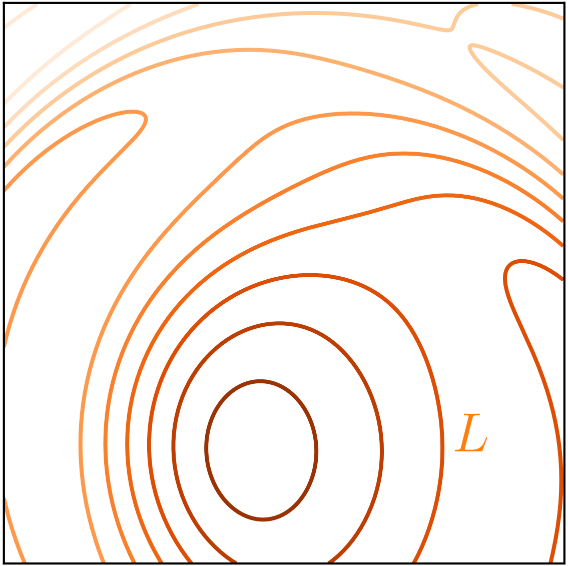
- From UC Riverside
  - CS 171 / EE 142: Introduction to Machine Learning and Data Mining
  - Professor Christian Shelton
- DO NOT REDISTRIBUTE
  - For use only by enrolled students in the course

Equality-Constrained Optimization

$$\operatorname*{arg\,min}_{\vec{z}} \quad L(\vec{z})$$

Equality-Constrained Optimization

$$\arg\min_{\vec{z}} \quad L(\vec{z})$$



Equality-Constrained Optimization

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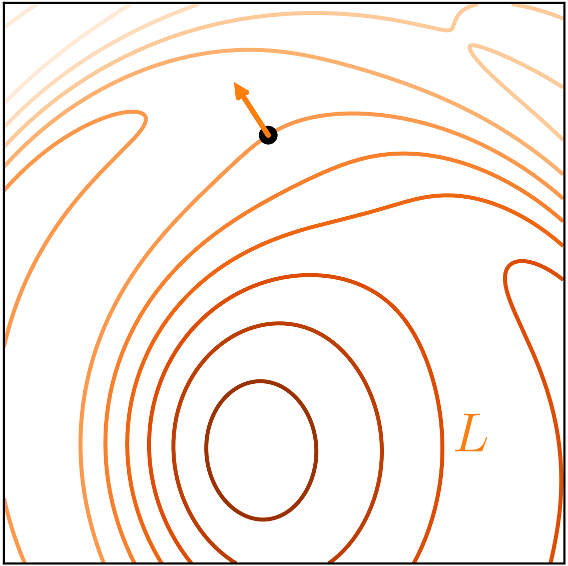

Equality-Constrained Optimization

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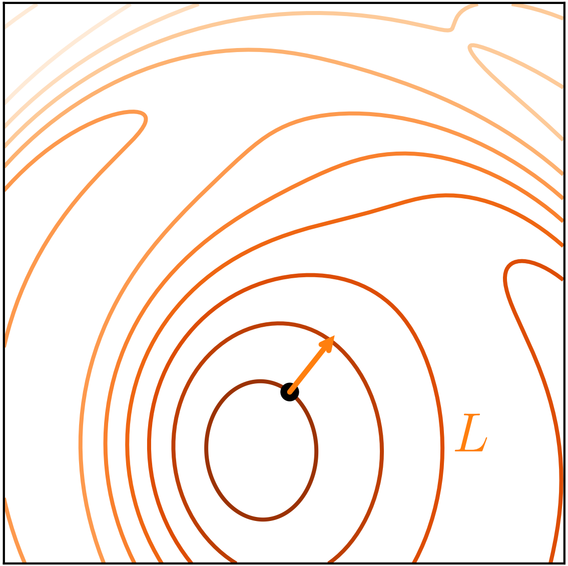
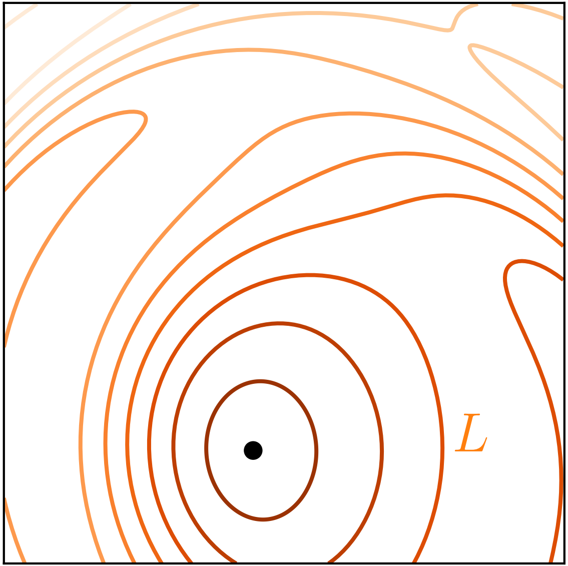


Equality-Constrained Optimization

$$\arg\min_{\vec{z}} \quad L(\vec{z})$$

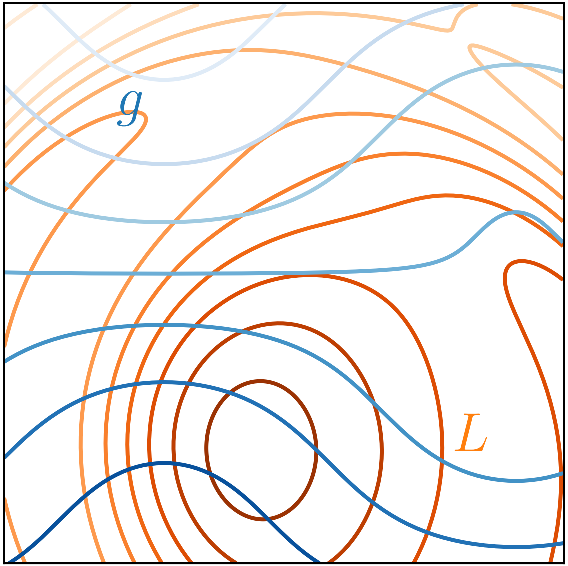
has necessary conditions of

$$\nabla L(\vec{z}) = \vec{0}$$



Equality-Constrained Optimization

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g(\vec{z}) = 0 \end{aligned}$$



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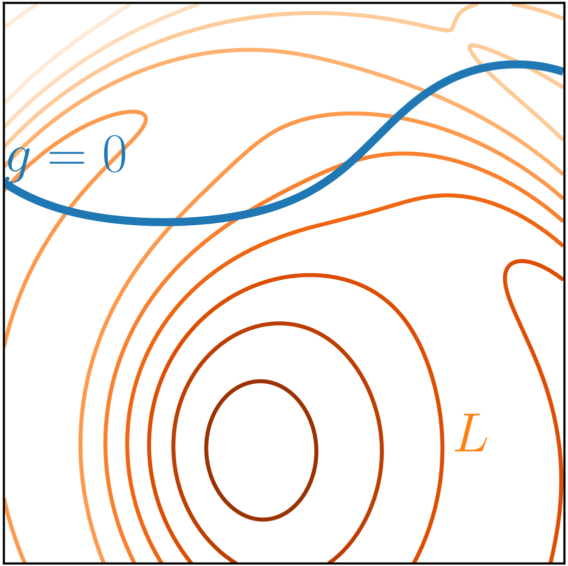

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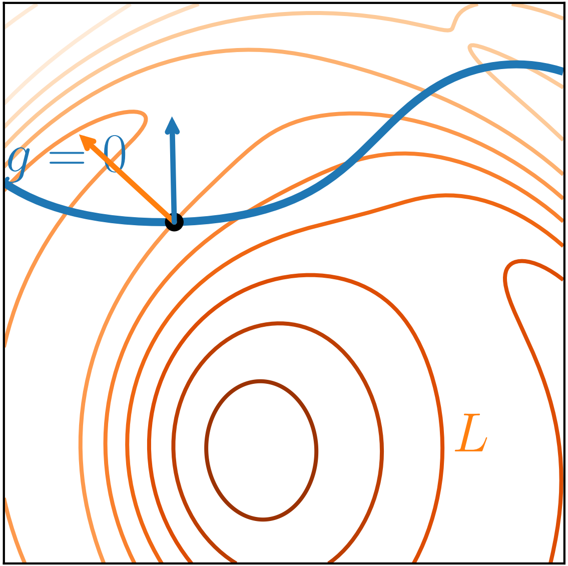

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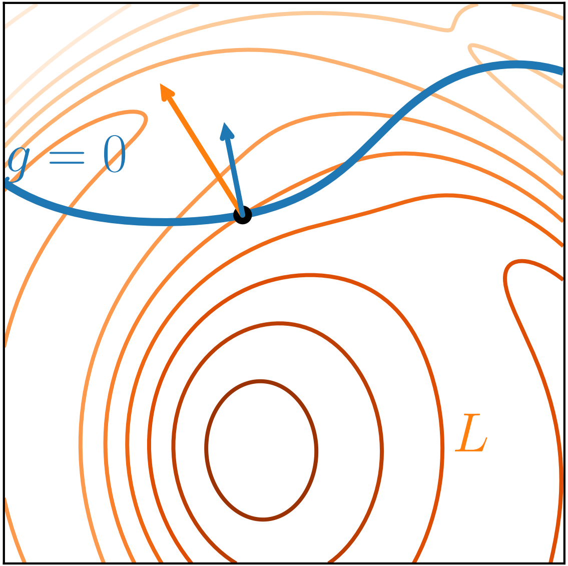
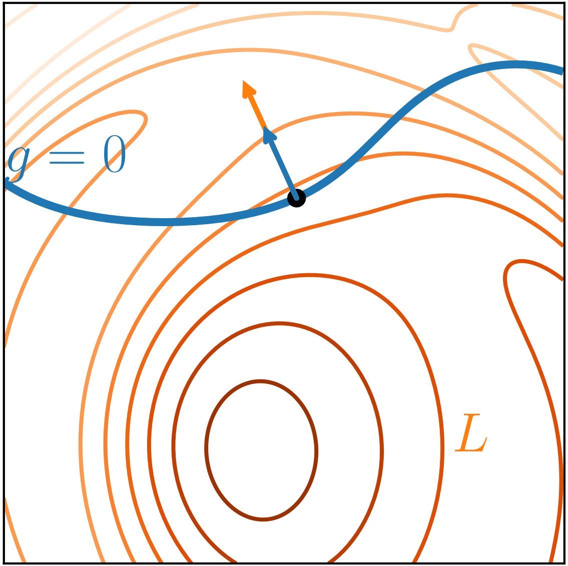


Equality-Constrained Optimization

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g(\vec{z}) = 0 \end{aligned}$$

has necessary conditions of

$$\begin{aligned} \nabla L(\vec{z}) &= \lambda \nabla g(\vec{z}) \\ g(\vec{z}) &= 0 \end{aligned}$$

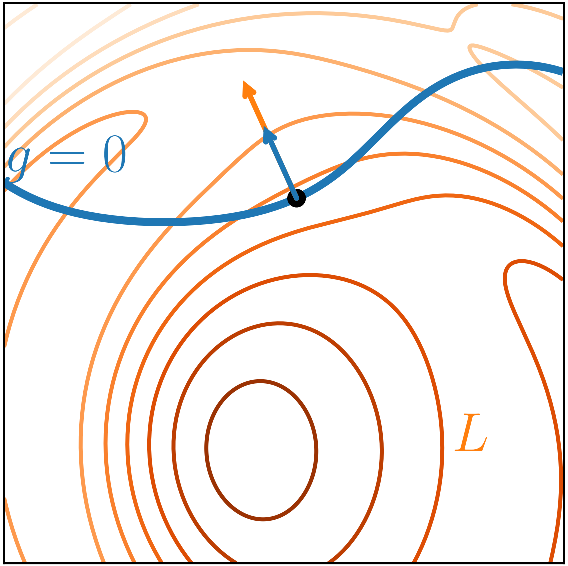


Equality-Constrained Optimization

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g(\vec{z}) = 0 \end{aligned}$$

has necessary conditions of

$$\begin{aligned} \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) &= \vec{0} \\ g(\vec{z}) &= 0 \end{aligned}$$



Equality-Constrained Optimization

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g(\vec{z}) = 0 \end{aligned}$$

has necessary conditions of

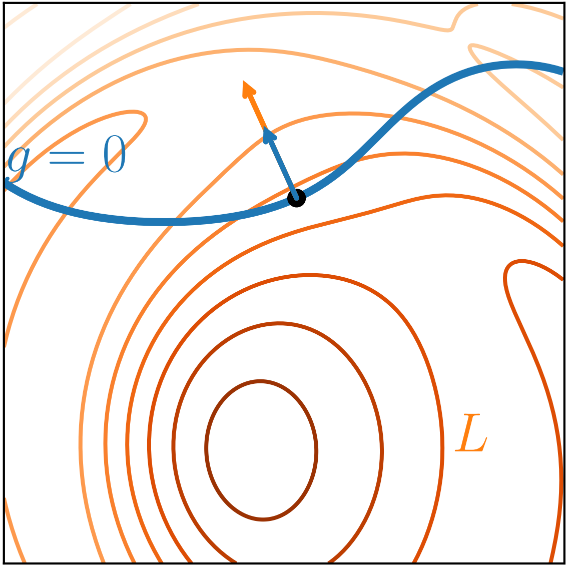
$$\begin{aligned} \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) &= \vec{0} \\ g(\vec{z}) &= 0 \end{aligned}$$

which implies the Lagrangian of

$$J = L(\vec{z}) - \lambda g(\vec{z})$$

because

$$\begin{aligned} \nabla_{\vec{z}} J = \vec{0} \quad & \rightarrow \quad \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) = \vec{0} \\ \frac{\partial}{\partial \lambda} J = 0 \quad & \rightarrow \quad g(\vec{z}) = 0 \end{aligned}$$



Equality-Constrained Optimization

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which implies the Lagrangian of

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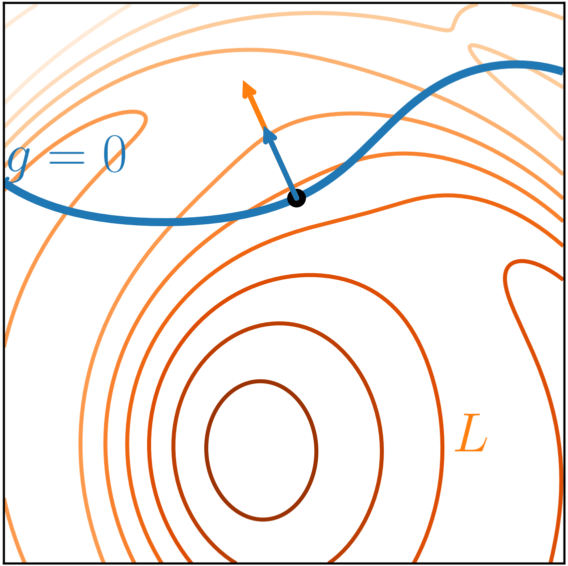
Lagrange multiplier

because

$$\begin{aligned} \nabla_{\vec{z}} J = \vec{0} \quad &\rightarrow \quad \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) = \vec{0} \\ \frac{\partial}{\partial \lambda} J = 0 \quad &\rightarrow \quad g(\vec{z}) = 0 \end{aligned}$$

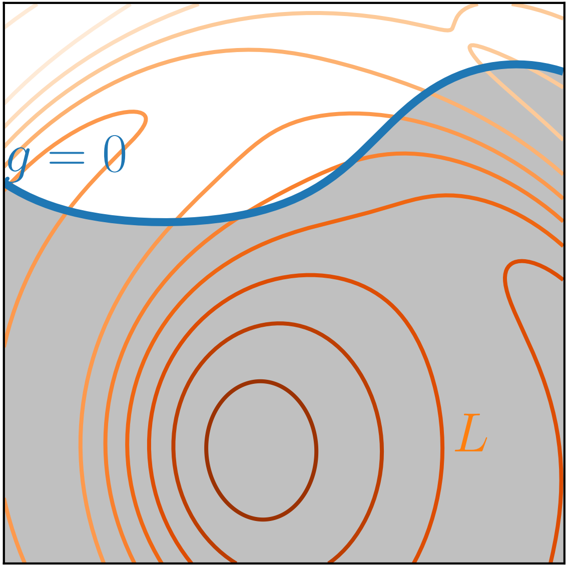


Inequality-Constrained Optimization

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g(\vec{z}) \geq 0 \end{aligned}$$

has necessary conditions of

$$\begin{aligned} \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) &= \vec{0} \\ g(\vec{z}) &= 0 \end{aligned}$$



Inequality-Constrained Optimization

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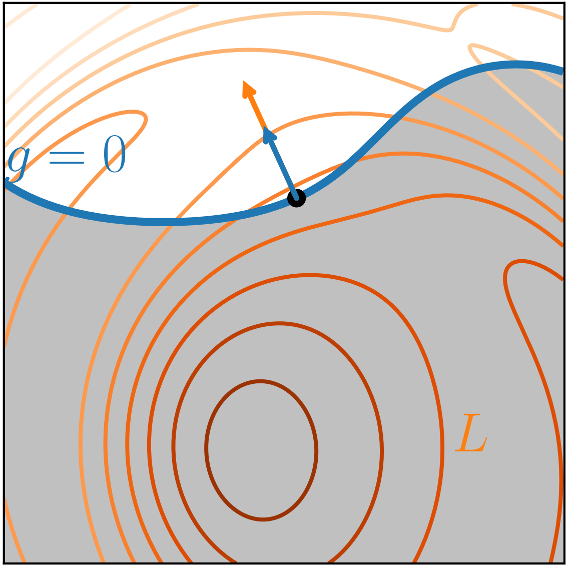

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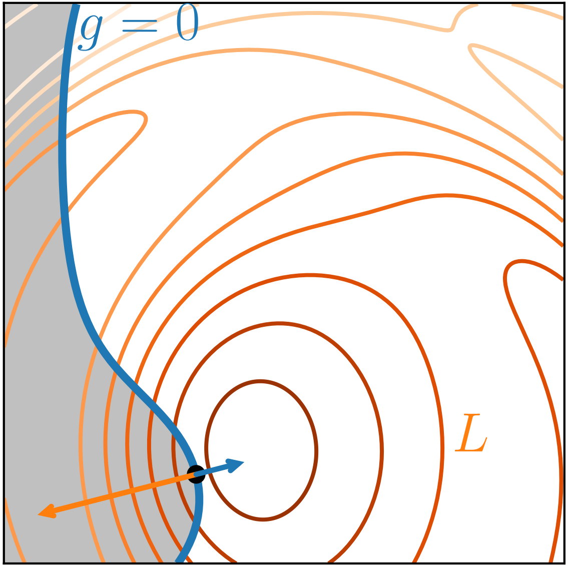

Inequality-Constrained Optimization

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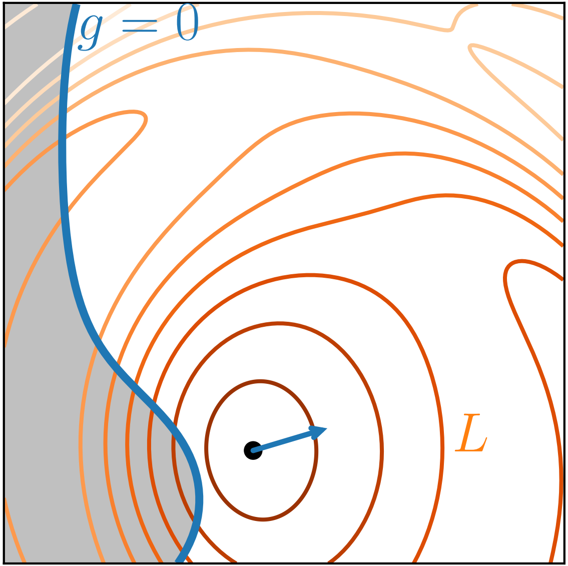
has necessary conditions of

$$\begin{aligned} \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) &= \vec{0} \\ g(\vec{z}) &= 0 \\ \lambda &\geq 0 \end{aligned}$$



Inequality-Constrained Optimization

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g(\vec{z}) \geq 0 \end{aligned}$$

has necessary conditions of

$$\begin{aligned} \nabla L(\vec{z}) - \lambda \nabla g(\vec{z}) &= \vec{0} \\ g(\vec{z}) &= 0 \\ \lambda &\geq 0 \end{aligned}$$

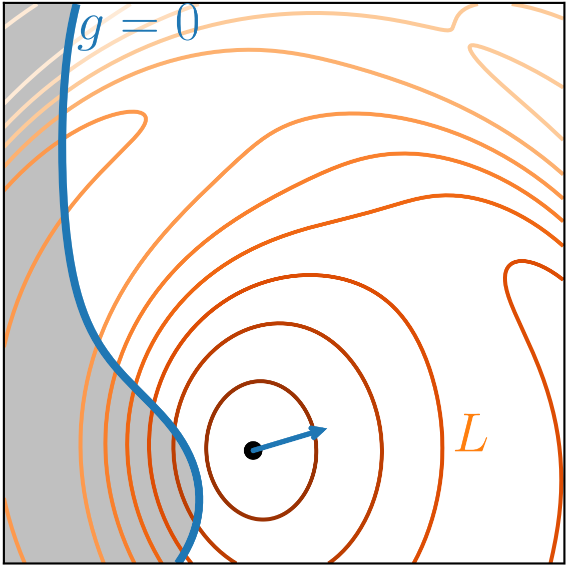
Therefore, we have

$$J = L(\vec{z}) - \lambda g(\vec{z})$$

plus the condition

$$\lambda \geq 0$$

If $\lambda = 0$, constraint is not active



General Lagrangians

Program

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g\ (\vec{z}) = 0 \end{aligned}$$

has the Lagrangian

$$J = L(\vec{z}) + \lambda\ g\ (\vec{z})$$

General Lagrangians

Program

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g_1(\vec{z}) = 0 \\ & g_2(\vec{z}) = 0 \\ & g_3(\vec{z}) = 0 \end{aligned}$$

has the Lagrangian

$$J = L(\vec{z}) + \lambda_1 g_1(\vec{z}) + \lambda_2 g_2(\vec{z}) + \lambda_3 g_3(\vec{z})$$

General Lagrangians

Program

$$\begin{aligned} \arg\min_{\vec{z}} \quad & L(\vec{z}) \\ \text{s.t.} \quad & g_1(\vec{z}) \geq 0 \\ & g_2(\vec{z}) \geq 0 \\ & g_3(\vec{z}) \geq 0 \end{aligned}$$

has the Lagrangian

$$J = L(\vec{z}) + \lambda_1 g_1(\vec{z}) + \lambda_2 g_2(\vec{z}) + \lambda_3 g_3(\vec{z})$$

with extra constraints

$$\lambda_1 \geq 0$$
$$\lambda_2 \geq 0$$
$$\lambda_3 \geq 0$$

SVM Lagrangian

(primal) QP:

$$
\begin{aligned}
\operatorname*{arg\,min}_{\vec{w},b,\xi_1,\ldots,\xi_m} \quad & \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i \\
\text{s.t.} \quad & y_i\left(\vec{x}_i^\top\vec{w} + b\right) \geq 1 - \xi_i \quad \text{for all } i \in \{1,\ldots,m\} \\
& \xi_i \geq 0 \quad \text{for all } i \in \{1,\ldots,m\}
\end{aligned}
$$

SVM Lagrangian

(primal) QP:

$$\begin{aligned} \operatorname*{arg\,min}_{\vec{w},b,\xi_1,\ldots,\xi_m} \quad & \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i \\ \text{s.t.} \quad & y_i\left(\vec{x}_i^\top\vec{w} + b\right) \geq 1 - \xi_i \quad && \text{for all } i \in \{1,\ldots,m\} \\ & \xi_i \geq 0 && \text{for all } i \in \{1,\ldots,m\} \end{aligned}$$

therefore has the following Lagrangian

$$J = \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m}a_i\left(y_i\left(\vec{x}_i^\top\vec{w} + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{m}r_i\xi_i$$

with constraints

$$\begin{aligned} a_i &\geq 0 \qquad && \text{for all } i \in \{1,\ldots,m\} \\ r_i &\geq 0 && \text{for all } i \in \{1,\ldots,m\} \end{aligned}$$

At the solution...

$$J = \frac{1}{2}\vec{w}^{\top}\vec{w} + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m} a_i\left(y_i\left(\vec{x}_i{}^{\top}\vec{w} + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{m} r_i\xi_i$$

$$\begin{aligned} 0 &= \nabla_{\vec{w}} J \\ &= \vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i \\ \vec{w} &= \sum_{i=1}^{m} a_i y_i \vec{x}_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial b} \\ &= \sum_{i=1}^{m} a_i y_i \\ 0 &= \sum_{i=1}^{m} a_i y_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial \xi_i} \\ &= C - a_i - r_i \\ r_i &= C - a_i \end{aligned}$$

At the solution...

$$J = \frac{1}{2}\vec{w}^{\intercal}\vec{w} + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m} a_i\left(y_i\left(\vec{x}_i^{\intercal}\vec{w} + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{m} r_i\xi_i$$

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$$J = \frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m} a_i\left(y_i\left(\vec{x}_i^{\mathsf{T}}\vec{w} + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{m}\left(C - a_i\right)\xi_i$$

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$$= \vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i$$
$$\vec{w} = \sum_{i=1}^{m} a_i y_i \vec{x}_i$$

$$0 = \frac{\partial J}{\partial b}$$
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$$0 = \sum_{i=1}^{m} a_i y_i$$

$$0 = \frac{\partial J}{\partial \xi_i}$$
$$= C - a_i - r_i$$
$$r_i = C - a_i$$

At the solution...

$$J = \frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m} a_i\left(y_i\left(\vec{x}_i^{\mathsf{T}}\vec{w} + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{m}\left(C - a_i\right)\xi_i$$

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At the solution...

$$J = \frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m} a_i\left(y_i\vec{x}_i^{\mathsf{T}}\vec{w} - 1 + \xi_i\right) - \sum_{i=1}^{m}\left(C - a_i\right)\xi_i$$

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$$J = \frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} - \sum_{i=1}^{m} a_i \left(y_i \vec{x}_i{}^{\mathsf{T}}\vec{w} - 1 + \xi_i\right) - \sum_{i=1}^{m} (-a_i)\, \xi_i$$

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At the solution...

$$J = \frac{1}{2}\vec{w}^\intercal \vec{w} - \sum_{i=1}^{m} a_i \left(y_i \vec{x}_i^\intercal \vec{w} - 1 \right)$$

$$\begin{aligned} 0 &= \nabla_{\vec{w}} J \\ &= \vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i \\ \vec{w} &= \sum_{i=1}^{m} a_i y_i \vec{x}_i \end{aligned}$$

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At the solution...

$$J = \frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i^{\mathsf{T}}\vec{w} + \sum_{i=1}^{m} a_i$$

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$$\begin{aligned} 0 &= \frac{\partial J}{\partial \xi_i} \\ &= C - a_i - r_i \\ r_i &= C - a_i \end{aligned}$$

At the solution...

$$J = -\frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} + \sum_{i=1}^{m} a_i$$

$$\begin{aligned} 0 &= \nabla_{\vec{w}} J \\ &= \vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i \\ \vec{w} &= \sum_{i=1}^{m} a_i y_i \vec{x}_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial b} \\ &= \sum_{i=1}^{m} a_i y_i \\ 0 &= \sum_{i=1}^{m} a_i y_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial \xi_i} \\ &= C - a_i - r_i \\ r_i &= C - a_i \end{aligned}$$

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$$J = -\frac{1}{2}\vec{w}^{\mathsf{T}}\vec{w} + \sum_{i=1}^{m} a_i$$

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$$\begin{aligned} 0 &= \frac{\partial J}{\partial b} \\ &= \sum_{i=1}^{m} a_i y_i \\ 0 &= \sum_{i=1}^{m} a_i y_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial \xi_i} \\ &= C - a_i - r_i \\ r_i &= C - a_i \end{aligned}$$

At the solution...

$$J = -\frac{1}{2}\left(\sum_{i=1}^{m} a_i y_i \vec{x}_i\right)^{\mathsf{T}} \left(\sum_{j=1}^{m} a_j y_j \vec{x}_j\right) + \sum_{i=1}^{m} a_i$$

$$\begin{aligned} 0 &= \nabla_{\vec{w}} J \\ &= \vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i \\ \vec{w} &= \sum_{i=1}^{m} a_i y_i \vec{x}_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial b} \\ &= \sum_{i=1}^{m} a_i y_i \\ 0 &= \sum_{i=1}^{m} a_i y_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial \xi_i} \\ &= C - a_i - r_i \\ r_i &= C - a_i \end{aligned}$$

At the solution...

$$J = -\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} a_i y_i \vec{x}_i^\intercal \vec{x}_j y_j a_j + \sum_{i=1}^{m} a_i$$

$$\begin{aligned} 0 &= \nabla_{\vec{w}} J \\ &= \vec{w} - \sum_{i=1}^{m} a_i y_i \vec{x}_i \\ \vec{w} &= \sum_{i=1}^{m} a_i y_i \vec{x}_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial b} \\ &= \sum_{i=1}^{m} a_i y_i \\ 0 &= \sum_{i=1}^{m} a_i y_i \end{aligned}$$

$$\begin{aligned} 0 &= \frac{\partial J}{\partial \xi_i} \\ &= C - a_i - r_i \\ r_i &= C - a_i \end{aligned}$$

At the solution...

$$
\begin{aligned}
J &= -\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} a_i y_i \vec{x}_i{}^{\intercal} \vec{x}_j y_j a_j + \sum_{i=1}^{m} a_i \\
&= -\frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} a_i G_{ij} a_j + \sum_{i=1}^{m} a_i \\
&= -\frac{1}{2}\vec{a}^{\intercal}\boldsymbol{G}\vec{a} + \vec{a}^{\intercal}\vec{1}
\end{aligned}
$$

where

$$
G_{ij} = y_i \vec{x}_i{}^{\intercal} \vec{x}_j y_j
$$

or

$$
\boldsymbol{G} = \begin{bmatrix} \text{---} y_1 \vec{x}_1^{\intercal} \text{---} \\ \text{---} y_2 \vec{x}_2^{\intercal} \text{---} \\ \vdots \\ \text{---} y_m \vec{x}_m^{\intercal} \text{---} \end{bmatrix} \begin{bmatrix} | & | & & | \\ y_1 \vec{x}_1 & y_2 \vec{x}_2 & \cdots & y_m \vec{x}_m \\ | & | & & | \end{bmatrix}^{\intercal}
$$

Dual SVM QP

$$J = -\frac{1}{2}\vec{a}^\top \boldsymbol{G}\vec{a} + \vec{a}^\top \vec{1}$$

$$\sum_{i=1}^{m} a_i y_i = 0$$

$$a_i \geq 0 \qquad \text{for all } i \in \{1, \ldots, m\}$$

$$r_i \geq 0 \qquad \text{for all } i \in \{1, \ldots, m\}$$

$$r_i = C - a_i \qquad \text{for all } i \in \{1, \ldots, m\}$$

Dual SVM QP

$$J = -\frac{1}{2}\vec{a}^{\mathsf{T}}\boldsymbol{G}\vec{a} + \vec{a}^{\mathsf{T}}\vec{1}$$

$$\vec{a}^{\mathsf{T}}\boldsymbol{Y} = 0$$

$$a_i \geq 0 \qquad \text{for all } i \in \{1, \ldots, m\}$$

$$r_i \geq 0 \qquad \text{for all } i \in \{1, \ldots, m\}$$

$$r_i = C - a_i \qquad \text{for all } i \in \{1, \ldots, m\}$$

Dual SVM QP

$$J = -\frac{1}{2}\vec{a}^{\mathsf{T}}\boldsymbol{G}\vec{a} + \vec{a}^{\mathsf{T}}\vec{1}$$

$$\vec{a}^{\mathsf{T}}\boldsymbol{Y} = 0$$

$$a_i \geq 0 \qquad \text{for all } i \in \{1, \ldots, m\}$$

$$a_i \leq C \qquad \text{for all } i \in \{1, \ldots, m\}$$

Dual SVM QP

$$J = -\frac{1}{2}\vec{a}^\top \boldsymbol{G}\vec{a} + \vec{a}^\top \vec{1}$$

$$\vec{a}^\top \boldsymbol{Y} = 0$$

$$0 \le a_i \le C \qquad \text{for all } i \in \{1, \ldots, m\}$$

Dual SVM QP

$$J = -\frac{1}{2}\vec{a}^\top \boldsymbol{G}\vec{a} + \vec{a}^\top \vec{1}$$

$$\vec{a}^\top \boldsymbol{Y} = 0$$

$$0 \leq a_i \leq C \qquad \text{for all } i \in \{1, \ldots, m\}$$

The Lagrangian should be *maximized* with respect to the Lagrange multipliers

dual QP:

$$\begin{aligned} \arg\max_{\vec{a}} \quad & -\frac{1}{2}\vec{a}^\top \boldsymbol{G}\vec{a} + \vec{a}^\top \vec{1} \\ \text{s.t.} \quad & 0 \leq a_i \leq C \qquad \text{for all } i \in \{1, \ldots, m\} \\ & \vec{a}^\top \boldsymbol{Y} = 0 \end{aligned}$$

Dual SVM QP

$$J = -\frac{1}{2}\vec{a}^{\mathsf{T}}\boldsymbol{G}\vec{a} + \vec{a}^{\mathsf{T}}\vec{1}$$

$$\vec{a}^{\mathsf{T}}\boldsymbol{Y} = 0$$

$$0 \leq a_i \leq C \qquad \text{for all } i \in \{1, \ldots, m\}$$

The Lagrangian should be *maximized* with respect to the Lagrange multipliers

dual QP:

$$\begin{aligned} \arg\min_{\vec{a}} \quad & \frac{1}{2}\vec{a}^{\mathsf{T}}\boldsymbol{G}\vec{a} - \vec{a}^{\mathsf{T}}\vec{1} \\ \text{s.t.} \quad & 0 \leq a_i \leq C \qquad \text{for all } i \in \{1, \ldots, m\} \\ & \vec{a}^{\mathsf{T}}\boldsymbol{Y} = 0 \end{aligned}$$

SVM QPs

Primal QP:

$$\begin{aligned}
\operatorname*{arg\,min}_{\vec{w},b,\xi_1,\ldots,\xi_m} \quad & \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i \\
\text{s.t.} \quad & y_i\left(\vec{x}_i^{\,\top}\vec{w} + b\right) \geq 1 - \xi_i && \text{for all } i \\
& \xi_i \geq 0 && \text{for all } i
\end{aligned}$$

Dual QP:

$$\begin{aligned}
\operatorname*{arg\,min}_{a_1,\ldots,a_m} \quad & \frac{1}{2}\vec{a}^\top\boldsymbol{G}\vec{a} - \vec{a}^\top\vec{1} \\
\text{s.t.} \quad & 0 \leq a_i \leq C && \text{for all } i \\
& \vec{a}^\top\boldsymbol{Y} = 0
\end{aligned}$$

SVM QPs

Primal QP:

$$\begin{aligned} \operatorname*{arg\,min}_{\vec{w},b,\xi_1,\ldots,\xi_m} \quad & \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i \\ \text{s.t.} \quad & y_i\left(\vec{x}_i^\top\vec{w} + b\right) \geq 1 - \xi_i && \text{for all } i \\ & \xi_i \geq 0 && \text{for all } i \end{aligned}$$

\# variables:

\# constraints:

Dual QP:

$$\begin{aligned} \operatorname*{arg\,min}_{a_1,\ldots,a_m} \quad & \frac{1}{2}\vec{a}^\top\boldsymbol{G}\vec{a} - \vec{a}^\top\vec{1} \\ \text{s.t.} \quad & 0 \leq a_i \leq C && \text{for all } i \\ & \vec{a}^\top\boldsymbol{Y} = 0 \end{aligned}$$

\# variables:

\# constraints:

SVM QPs

Primal QP:

$$\begin{aligned} \operatorname*{arg\,min}_{\vec{w},b,\xi_1,\ldots,\xi_m} \quad & \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i \\ \text{s.t.} \quad & y_i\left(\vec{x}_i^\top\vec{w} + b\right) \geq 1 - \xi_i \quad \text{for all } i \\ & \xi_i \geq 0 \quad \text{for all } i \end{aligned}$$

\# variables: $n + m + 1$
\# constraints: $2m$

Dual QP:

$$\begin{aligned} \operatorname*{arg\,min}_{a_1,\ldots,a_m} \quad & \frac{1}{2}\vec{a}^\top\boldsymbol{G}\vec{a} - \vec{a}^\top\vec{1} \\ \text{s.t.} \quad & 0 \leq a_i \leq C \quad \text{for all } i \\ & \vec{a}^\top\boldsymbol{Y} = 0 \end{aligned}$$

\# variables: m
\# constraints: $m + 1$

SVM QPs

Primal QP:

$$\begin{aligned} \operatorname*{arg\,min}_{\vec{w},b,\xi_1,\ldots,\xi_m} \quad & \frac{1}{2}\vec{w}^\top\vec{w} + C\sum_{i=1}^{m}\xi_i \\ \text{s.t.} \quad & y_i\left(\vec{x}_i^{\top}\vec{w} + b\right) \geq 1 - \xi_i \quad \text{for all } i \\ & \xi_i \geq 0 \quad \text{for all } i \end{aligned}$$

\# variables: $n + m + 1$
\# constraints: $2m$

directly produces $\vec{w}$ and b

Dual QP:

$$\begin{aligned} \operatorname*{arg\,min}_{a_1,\ldots,a_m} \quad & \frac{1}{2}\vec{a}^\top \boldsymbol{G}\vec{a} - \vec{a}^\top\vec{1} \\ \text{s.t.} \quad & 0 \leq a_i \leq C \quad \text{for all } i \\ & \vec{a}^\top\boldsymbol{Y} = 0 \end{aligned}$$

\# variables: m
\# constraints: $m + 1$

$\vec{w} = \sum_{i=1}^{m} a_i y_i \vec{x}_i$
b is more complicated (see book)

Support Vectors

Recall Lagrangian has term

$$\overbrace{a_i}^{\text{Lagrange multipler}} \underbrace{\left(y_i \left(\vec{x}_i^{\mathsf{T}} \vec{w} + b\right) - 1 + \xi_i\right)}_{\text{constraint}}$$

Support Vectors

Recall Lagrangian has term

$$\overbrace{a_i}^{\text{Lagrange multipler}} \underbrace{\left(y_i\left(\vec{x}_i^{\top}\vec{w}+b\right)-1+\xi_i\right)}_{\text{constraint}}$$

Either

- Constraint is active:
  $y_i\left(\vec{x}_i^{\top}\vec{w}+b\right)=1-\xi_i$
  [point is on or inside the margin], or
- Constraint is not active:
  $a_i=0$
  [point is outside the margin]

Points for which $a_i>0$ are <u>support vectors</u>

$$\vec{w}=\sum_{i=1}^{m} a_i y_i \vec{x}_i$$

Support Vectors

Recall Lagrangian has term

$$\overbrace{a_i}^{\text{Lagrange multipler}} \underbrace{\left(y_i\left(\vec{x}_i^{\mathsf{T}} \vec{w} + b\right) - 1 + \xi_i\right)}_{\text{constraint}}$$

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- Constraint is not active:
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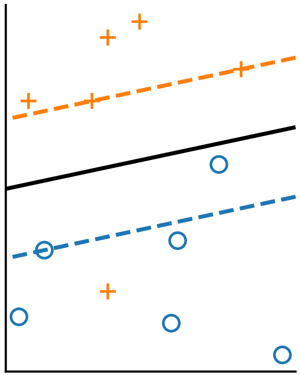
  [point is outside the margin]

Points for which $a_i > 0$ are support vectors

$$\vec{w} = \sum_{i=1}^{m} a_i y_i \vec{x}_i$$



Support Vectors

Recall Lagrangian has term

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